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Roll No

FT-7003-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Heat Transfer

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Derive steady state heat conduction in a plane wall.
b) Explain critical thickness of insulation for cylinder with figure.
2. a) Explain heat transfer in boundary layer concept.
b) Define contact resistance and overall heat transfer coefficient.
3. a) Derive expression for temperature distribution, under one dimensional steady state heat conduction for composite wall.
b) An exterior wall of a house may be approximated by a 0.1 m layer of common brick ($k = 0.7 \text{ W/m}^\circ\text{C}$) followed by 0.04m layer of gypsum plaster ($k=0.48 \text{ W/m}^\circ\text{C}$). What thickness of loosely packed rock wool insulation ($k=0.065 \text{ W/m}^\circ\text{C}$) should be added to reduce the heat loss or gain through the wall by 80%.
4. a) What is the difference between natural convection and forced convection.

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- b) The velocity distribution in the boundary layer is given by

$$\frac{u}{v} = \frac{3}{2} \frac{y}{\delta} - \frac{1}{2} \frac{y^2}{\delta^2}, \delta \text{ being boundary layer thickness.}$$

Calculate the following-

- i) Ratio of displacement thickness to boundary layer

$$\text{thickness} \left(\frac{\delta^*}{\delta} \right)$$

- ii) Ratio of momentum thickness to boundary layer

$$\text{thickness} \left(\frac{\theta}{\delta} \right).$$

5. a) Derive the following relation for the radiant heat exchange between two gray surface, using definition of irradiation and radiosity

$$(Q_{12})_{net} = \frac{A_1 \sigma (T_1^4 - T_2^4)}{\frac{1 - \epsilon_1}{\epsilon_1} + \frac{1}{F_{1-2}} + \left(\frac{1 - \epsilon_2}{\epsilon_2} \right) \frac{A_1}{A_2}}$$

- b) Explain evaporation. What are various types of evaporator? Explain briefly.
6. Define Kirchhoff Law Planck's and emissivity.
7. Describe the various steps involved in single effect evaporator calculation.
8. a) Explain schematic representations of flow configurations for multipass shell and tube heat exchangers.
- b) Water at the rate of 68 kg/min is heated from 35 to 75°C. By an oil having a specific heat of 1.9KJ/Kg°C. The fluids are used in a counterflow double pipe heat exchanger, and the oil enters the exchanger at 110°C and leaves at 75°C. The overall heat transfer coefficient is 320W/m²°C. Calculate the heat exchanger area.

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